InjectorWIZARD - Information

Author: Sam Owen

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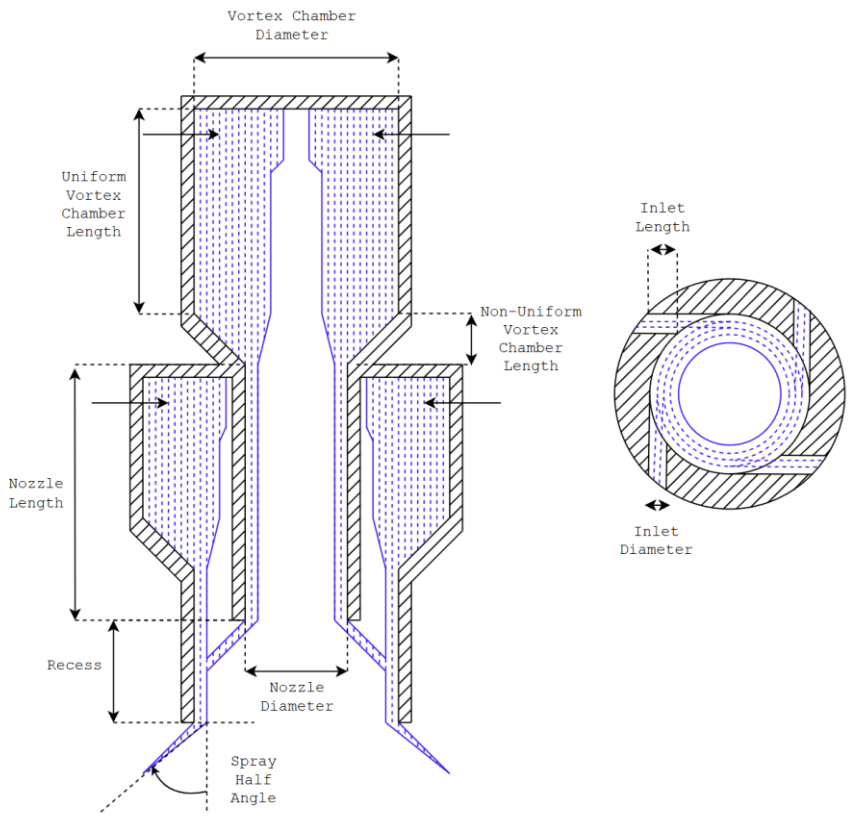
1. Swirl and Jet Injectors [1]



The filling efficiency of the swirl injector is defined as the ratio of the area filled by fluid to the total area.

$$\varphi = 1 - \frac{r_N^2}{R_N^2} = 1 - \hat{r}_N^2$$

The geometric constant is another important parameter which relates injector geometries. This is defined as follows.

$$A \equiv \frac{R_N(R_V - R_T)}{nR_T^2} = \frac{(1-\varphi)\sqrt{2}}{\varphi\sqrt{\varphi}}$$

The filling efficiency can then be used to calculate the mass flow coefficient of the injector.

$$\mu = \sqrt{\frac{\varphi^3}{2-\varphi}}$$

The ideal total velocity magnitude in the injector can be determined from potential flow. This is given by the following equation.

$$U_\Sigma = \sqrt{\frac{2\Delta P_\Sigma}{\rho}}$$

The nozzle radius can then be determined from these parameters for a given filling efficiency.

$$R_N = \sqrt{\frac{\dot{m}_\Sigma}{\pi\mu\sqrt{2\rho\Delta P_\Sigma}}}$$

The coefficient of nozzle opening is defined as follows for the injector.

$$\eta = \frac{R_V - R_T}{R_N}$$

From the definition of the geometric constant and coefficient of nozzle opening, the remaining geometries can be calculated.

$$R_T = \sqrt{\frac{\eta R_N^2}{nA}}$$

$$R_V = \eta R_N + R_T$$

The pressure drop across the tangential inlets can be expressed as a function of the total pressure drop across the injector. This is valid only for coefficient of nozzle opening values greater than one.

$$\Delta P_T = \Delta P_\Sigma \frac{\hat{r}_{V0}^2}{\eta^2}$$

The ideal inflow velocity to the vortex chamber, which is equivalent to the velocity in the tangential inlets, is then given by this equation.

$$U_{in} = \sqrt{\frac{2\Delta P_T}{\rho}}$$

The spray half angle downstream of the injector exit can then be found from the following relationship. The velocity components vary within the liquid film, so there is some uncertainty in this value.

$$\tan\frac{\alpha}{2} = \frac{\mu A}{\sqrt{1 - \mu^2 A^2}}$$

The continuity and momentum equations for the swirl injector are used to determine the film radius in the vortex chamber. Conservation of angular momentum yields the following relationship.

$$U_{j,\theta} = U_{\Sigma}\frac{\hat{r}_{V0}}{\hat{r}_j}$$

The axial velocity component can be found using the following equation.

$$U_{j,z} = U_{\Sigma}\sqrt{1 - \frac{\hat{r}_{V0}^2}{\hat{r}_j^2}}$$

This is equivalent to the flow continuity equation, which can be expressed as follows.

$$U_{j,z}\left(\hat{R}_j^2 - \hat{r}_j^2\right) = U_{N,z}\left(\hat{R}_N^2 - \hat{r}_N^2\right)$$

The corresponding dimensionless film radius at a given injector stage is given by the following equation.

$$\hat{r}_j = \sqrt{1 - \varphi_j}$$

Where the effective filling efficiencies at the nozzle and head-end stages are defined as follows.

$$\varphi_N = \varphi$$

$$\varphi_{V0} = \frac{3\varphi - 2\varphi^2}{2 - \varphi}$$

The internal mixing recess of coaxial elements is found using the following equation in terms of the propellant ratio and the mixing time. The mixing time is set as 0.2ms for non-hypergolic propellants and 0.1ms for hypergolic propellants.

$$l_{mix} = \tau\sqrt{2}\left(\frac{K_m\mu_2}{(K_m + 1)\varphi_2}\sqrt{\frac{\Delta P_{\Sigma,2}}{\rho_2}} + \frac{\mu_1}{(K_m + 1)\varphi_1}\sqrt{\frac{\Delta P_{\Sigma,1}}{\rho_1}}\right)$$

The mixing spray angle of the coaxial elements is then calculated assuming a full momentum transfer between the inner and outer elements. In the jet centred swirl configuration, the jet orifice is sized according to the standard orifice equation.

$$R = \sqrt{\frac{\dot{m}}{\pi C_D\sqrt{2\rho\Delta P}}}$$

In the swirl centred jet configuration, the annular jet is sized according to the following equation.

$$R = \sqrt{\frac{\dot{m}}{\pi C_D \sqrt{2\rho\Delta P}} + (R_N + \delta)^2}$$

This assumes that the jet exit is adjacent to the outer wall of the swirl injector nozzle. For swirl injector dynamics, the following linear theory is used. The complex response function of the liquid injector is given by the following equation. Distances are normalized with respect to the nozzle radius and velocities with respect to the inflow velocity.

$$\Pi_\Sigma = \frac{\dot{m}'_N / \dot{m}_N}{\Delta P'_\Sigma / \Delta P_\Sigma} = \left(\frac{\hat{R}_V}{\hat{r}_{V0}}\right)^2 \frac{\Pi_T \Pi_{VN} \Pi_N}{1 + 2\Pi_T(\Pi_{V2} + \Pi_{V3})}$$

This function quantifies the injector's sensitivity to upstream perturbations, and each component represents a different stage of the injector flow path. The complex response of the tangential inlets to plane waves is defined as follows.

$$\Pi_T = \frac{\dot{m}'_T / \dot{m}_T}{\Delta P'_T / \Delta P_T}$$

The complex response of the vortex chamber to surface waves is characterized by reflections at the boundary between the vortex chamber and nozzle.

$$\Pi_{VN} = \frac{\dot{m}'_{VN} / \dot{m}_{VN}}{\dot{m}'_T / \dot{m}_T}$$

The complex response of the nozzle to surface waves is defined from the proportion of transmitted waves through the reflective boundary.

$$\Pi_N = \frac{\dot{m}'_N / \dot{m}_N}{\dot{m}'_{VN} / \dot{m}_{VN}}$$

The complex response functions of the closed end of the vortex chamber comprise a feedback connection and are associated with surface waves and vorticity waves respectively.

$$\Pi_{V2} = \frac{1}{2}\frac{\Delta P'_{V2}/\Delta P_T}{\dot{m}'_T/\dot{m}_T}$$

$$\Pi_{V3} = \frac{1}{2}\frac{\Delta P'_{V3}/\Delta P_T}{\dot{m}'_T/\dot{m}_T}$$

The injector response function and its component parts are defined as follows. The response of the tangential inlets to pressure oscillations is given by the following equation in terms of the Strouhal number at this section of the injector.

$$\Pi_T = \frac{1 - iSh_T}{2 + 2Sh_T^2}$$

$$Sh_T = \hat{\omega}\frac{\hat{L}_T}{\hat{U}_T}$$

The phase shift for the vortex chamber is defined by the following equation.

$$\Phi_V = \hat{\omega}\frac{\hat{L}_V + \hat{L}_C}{\hat{c}_V}$$

The liquid wave speed in the uniform section is given by the following equation related to the angular momentum constant of the injector.

$$\hat{c}_V = \hat{U}_{V,z} \pm \sqrt{\hat{C}^2\frac{\hat{R}_V^2 - \hat{r}_V^2}{2\hat{r}_V^4}}$$

The nozzle reflection coefficient is defined by the following expression. This is determined assuming that waves are reflected at an approximated sudden-step discontinuity between the vortex chamber and the nozzle.

$$\Pi_{refl} = 1 - 2\sqrt{\frac{\varphi}{\hat{R}_V^2 - \hat{r}_{V0}^2}}$$

The transfer function of surface waves in the vortex chamber is then defined as follows.

$$\Pi_{V2} = \frac{1}{A\sqrt{2\frac{\left(\hat{R}_V^2 - \hat{r}_{V0}^2\right)}{\hat{R}_N^2}}}\sum_{n=0}^{\infty}\Pi_{refl}^n e^{i(-n\Phi_{VD}+n\Phi_{VU})-\nu(n\Phi_{VD}+n\Phi_{VU})}$$

The transfer function of the nozzle entrance is given by this equation. An artificial viscosity coefficient is used to dampen infinite reflections.

$$\Pi_{VN} = \sum_{n=0}^{\infty} \Pi_{refl}^{n} e^{i(-(n+1)\Phi_{VD}+n\Phi_{VU})-\nu((n+1)\Phi_{VD}+n\Phi_{VU})}$$

The Strouhal number at the vortex chamber is given by this equation.

$$Sh_V = \hat{\omega}\frac{\hat{R}_V}{\hat{U}_{V,z}}$$

A new function and integration constant are introduced for analysing this stage.

$$f(\bar{x}) = \bar{x}Sh_V \tan\left(\frac{\pi\bar{x}}{2}\right)$$

$$K = 1 - \frac{\hat{r}_{V0}}{\hat{R}_V}$$

The real and imaginary parts of the vorticity wave transfer function are given by the two following equations.

$$Re(\Pi_{V3}) = 2\int_0^1 \frac{\cos(f(\bar{x}))}{(1-K\bar{x})^3} e^{-\nu f(\bar{x})}\, d\bar{x}$$

$$Im(\Pi_{V3}) = -2\int_0^1 \frac{\sin(f(\bar{x}))}{(1-K\bar{x})^3} e^{-\nu f(\bar{x})}\, d\bar{x}$$

The phase shift for the nozzle is defined by the following equation.

$$\Phi_N = \hat{\omega}\frac{\hat{L}_N}{\hat{c}_N}$$

The liquid wave speed in the nozzle is given by the following expression.

$$\hat{c}_N = \hat{U}_{N,z} + \sqrt{\hat{C}^2 \frac{\hat{R}_N^2 - \hat{r}_N^2}{2\hat{r}_N^4}}$$

The nozzle transfer function is then defined by the following relationship which is based on the proportion of transmitted waves through the reflective boundary.

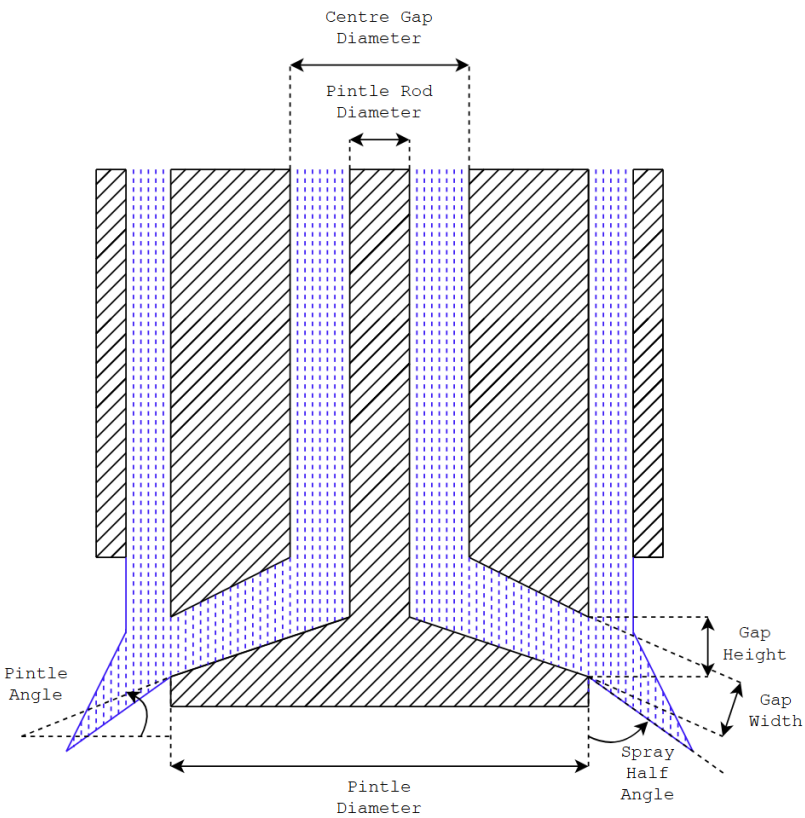
$$\Pi_N = (1 - \Pi_{refl})e^{-i\Phi_N}$$

The concluding step in this analysis is to combine all transfer functions into the total injector response function.

$$\Pi_\Sigma = \frac{\dot{m}'_N / \dot{m}_N}{\Delta P'_\Sigma / \Delta P_\Sigma} = \left(\frac{\hat{R}_V}{\hat{r}_{V0}}\right)^2 \frac{\Pi_T \Pi_{VN} \Pi_N}{1 + 2\Pi_T(\Pi_{V2} + \Pi_{V3})}$$

Open swirl injectors have a vortex chamber radius equal to the nozzle radius and therefore tend to have a more stable dynamic response.

2. Pintle Injectors [2,3]



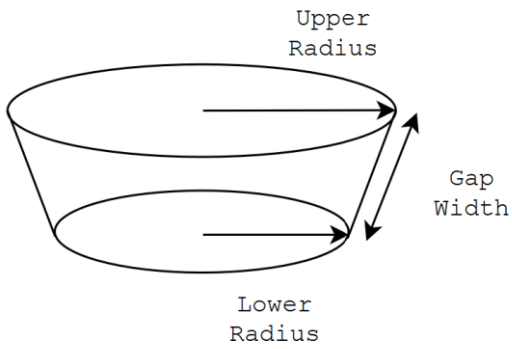
Based on the input for maximum mass flow rate, the pintle area is calculated using the orifice equation.

$$A = \frac{\dot{m}}{C_D\sqrt{2\rho\Delta P}}$$

The minimum centre gap diameter required to provide the input flow rate is then given by the following equation.

$$D_{CG} = 2\sqrt{\frac{A}{\pi} + \left(\frac{D_{PR}}{2}\right)^2}$$

The pintle exit area is analogous to the surface area of a frustrum. In this case, the upper radius is equal to the pintle radius, and the lower radius can be found geometrically based on the pintle angle and the gap width.



The pintle exit area is then given by the following expression in terms of gap width.

$$A_P = \pi(2R_P - w\sin\theta)w$$

In throttling systems, the gap height is used as a tracking distance.

$$w = h\cos\theta$$

The spray angle is then determined from the momentum balance of the colliding streams. The fixed pintle configuration replaces the continuous gap with a series of holes. The radius of a single hole is calculated from the total orifice area of the pintle, the number of holes per layer, and the number of layers.

$$r = \sqrt{\frac{A}{\pi n N}}$$

The minimum pintle radius required to fit a certain number of holes in a layer is the circumradius of a regular polygon with side length equal to the diameter of each hole.

$$R_{P,min} = \frac{r}{\sin\frac{\pi}{n}}$$

The collision area between the stream of one pintle hole and the annular element stream is given by the following equation.

$$A_{collision} = 2r\left(\sqrt{R_A^2 - r^2} - \sqrt{(R_P + \delta)^2 - r^2}\right) - \frac{1}{2}(R_P + \delta)^2(\beta - \sin\beta) + \frac{1}{2}R_A^2(\gamma - \sin\gamma)$$

Where the segment angles are evaluated in radians using the following equations.

$$\beta = 2\arcsin\frac{r}{R_P + \delta}$$

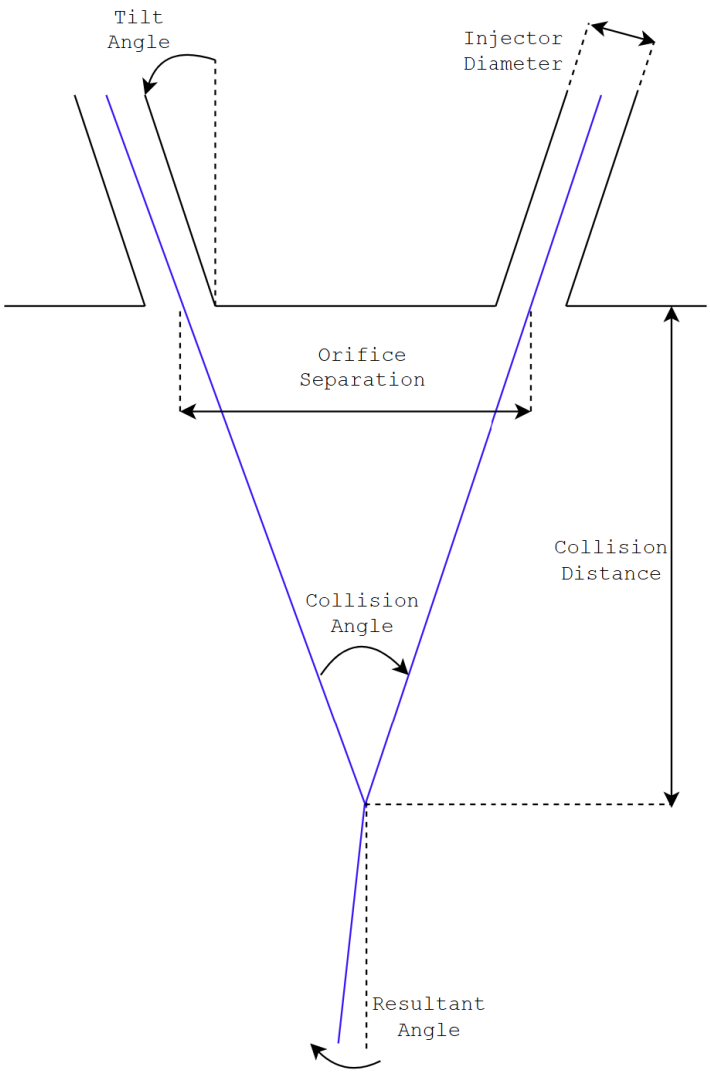
$$\gamma = 2\arcsin\frac{r}{R_A}$$

The collision efficiency is then defined as the ratio of the layer collision area to the full area of the annular stream.

$$\eta = \frac{A_{collision} n}{\pi R_A^2 - \pi(R_P + \delta)^2}$$

The spray angle of the fixed pintle injector is then determined from the momentum balance of the colliding areas of the pintle layer and annular stream.

3. Impinging Injectors [2,4]



The size of each injector is calculated using the orifice equation.

$$R = \sqrt{\frac{\dot{m}}{\pi C_D \sqrt{2\rho\Delta P}}}$$

The resultant angle is then found from the momentum balance of the colliding streams.

$$\beta = \arctan \frac{\dot{m}_1 U_1 \sin\alpha_1 - \dot{m}_2 U_2 \sin\alpha_2}{\dot{m}_1 U_1 \cos\alpha_1 + \dot{m}_2 U_2 \cos\alpha_2}$$

The triplet configuration imposes the condition that the centred element is not angled, and so the resultant angle is found the momentum balance between the two angled elements. The collision distance ratio is then defined as the ratio between the collision distance and the average orifice diameter.

4. Combustion Chamber [5]

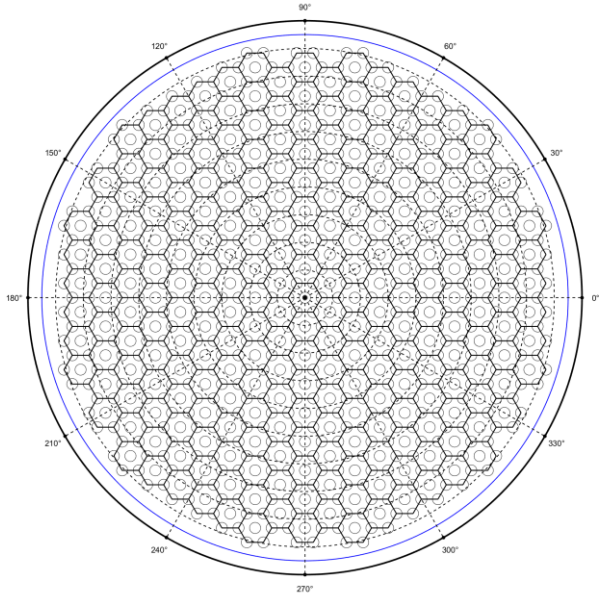
The resonant modes of the combustion chamber are calculated using the following equation related to the chamber geometry, the local speed of sound, and roots of the Bessel function of the first kind describing the system.

$$f_{m,n,q} = \frac{a}{2}\sqrt{\left(\frac{\alpha_{mn}}{R}\right)^2 + \left(\frac{q}{L}\right)^2}$$

Here m, n, and q are wavenumbers (zero or integer) which correspond to radial, tangential, and longitudinal oscillations.

α_{mn}		m				
		0	1	2	3	4
n	0	0.000	1.220	2.333	3.238	4.241
	1	0.568	1.697	2.717	3.726	4.731
	2	0.972	2.135	3.173	4.192	5.204
	3	1.337	2.551	3.612	4.643	5.662
	4	1.693	2.995	4.037	5.082	6.110

5. Pattern



The distance between lattice points can be found from the element spacing and the element radius.

$$\lambda = 2r + \mu$$

Each hexagonal point can be defined by two integers which map to Cartesian coordinates using the following equations.

$$x = \frac{3}{2}\lambda q$$

$$y = \lambda\sqrt{3}\left(s + \frac{q}{2}\right)$$

Valid lattice sites are defined using the combustion chamber radius, element radius, outer exclusion distance, and inner exclusion distance.

$$\delta + r \leq \sqrt{x^2 + y^2} \leq R - r - \varepsilon$$

The infinite-lattice packing efficiency of separated elements is given by the following equation.

$$\eta_\infty = \frac{\pi}{2\sqrt{3}}\left(\frac{2r}{2r + \mu}\right)^2$$

The packaging efficiencies of the pattern are then given by these relationships.

$$\eta_T = N\left(\frac{r}{R}\right)^2$$

$$\eta_U = N\frac{r^2}{(R - \varepsilon)^2 - \delta^2}$$

References:

[1] V. Bazarov, V. Yang, *Design and Dynamics of Jet and Swirl Injectors*. AIAA (American Institute of Aeronautics & Astronautics), 2004.

[2] G. Sutton and O. Biblarz, *Rocket Propulsion Elements*, 7th ed. New York: John Wiley & Sons Inc, 2001.

[3] G. Dressler, J. Bauer, *TRW Pintle Engine Heritage and Performance Characteristics*. AIAA (American Institute of Aeronautics & Astronautics), 2000.

[4] D. Huzel, D. Huang. *Modern Engineering for Design of Liquid-Propellant Rocket Engines.* AIAA (American Institute of Aeronautics & Astronautics), 1992.

[5] M. Zucrow, J. Hoffman. *Gas Dynamics. Volume 2 – Multidimensional Flow.* New York: John Wiley & Sons Inc, 1977.